

Physics-Informed Learning vs MLP

Technical Report — Learning Methods

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Abstract

This report compares two data-driven approaches to inverse dynamics identification for a 2-DOF rotating parallelogram four-bar linkage under a time-varying external load with sensor latency. The first approach embeds the complete Lagrangian structure into the learning problem—fixing the analytical form of the inertia, Coriolis, and gravity matrices and optimising only five scalar physical constants. The second uses a standard three-layer multilayer perceptron (MLP) with no physics knowledge. With 1379 training samples (10% of the full dataset), the physics-informed model achieves an out-of-distribution torque RMSE of 0.029 N·m versus 0.919 N·m for the MLP, using $7000\times$ fewer parameters. In closed-loop Computed Torque Control (CTC) validation on a held-out trajectory, the physics-informed model tracks with 0.635 mm RMSE while the MLP diverges within 1.3 s.

1 Introduction

Accurate inverse dynamics models are essential for torque-based control of robotic mechanisms. The inverse dynamics problem asks: given a desired motion $(\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}})$ and external load $\mathbf{F}$, what joint torques $\boldsymbol{\tau}$ are required? Analytical models derived from Lagrangian mechanics are generalisable and physically consistent, but their derivation requires significant effort. Black-box function approximators such as MLPs avoid this effort but tend to overfit the training distribution and fail on states not seen during training.

Physics-informed learning offers a middle ground: the analytical structure of the model is fixed from first principles, and only the scalar physical constants are estimated from data. This retains the generalisation guarantees of the analytical model while enabling automatic parameter identification from measurements.

This report uses a parallelogram four-bar linkage mounted on a rotating base as the testbed. The Lagrangian dynamics of this mechanism are derived in the companion technical report [1], which shows that the equations of motion reduce to a structurally simple form parameterised by exactly five composite constants. These five constants are the quantities learned by the physics-informed model.

2 Mechanical System and Equations of Motion

The mechanism has two degrees of freedom in minimal coordinates $\mathbf{q} = [q_1, q_2]^\top$, where q_1 is the absolute rotation of the base arm and q_2 is the relative angle of the side links. The equations of motion in standard manipulator form are [1]

$$M(q_2) \ddot{\mathbf{q}} + C(q_2, \dot{\mathbf{q}}) \dot{\mathbf{q}} + G(\mathbf{q}) = \boldsymbol{\tau} + Q_F(\mathbf{q}, \mathbf{F}), \quad (1)$$

with the parallelogram constraint reducing all matrices to

$$M(q_2) = \begin{bmatrix} M_{11,0} + 2K_1 \cos q_2 & K_2 + K_1 \cos q_2 \\ K_2 + K_1 \cos q_2 & K_2 \end{bmatrix}, \quad (2)$$

$$(C\dot{\mathbf{q}})_1 = -K_1 \sin q_2 (2\dot{q}_1 \dot{q}_2 + \dot{q}_2^2), \quad (C\dot{\mathbf{q}})_2 = K_1 \sin q_2 \dot{q}_1^2, \quad (3)$$

$$G(\mathbf{q}) = g \begin{bmatrix} m_{\text{eff},1} \cos q_1 + m_{\text{eff},2} \cos(q_1 + q_2) \\ m_{\text{eff},2} \cos(q_1 + q_2) \end{bmatrix}. \quad (4)$$

The five composite constants $\{K_1, K_2, M_{11,0}, m_{\text{eff},1}, m_{\text{eff},2}\}$ are the only physically observable combinations from torque measurements. Individual body masses and moments of inertia cannot be recovered separately. Their ground-truth values, computed from the nominal physical parameters of the mechanism, are given in Table 1.

Table 1: Ground-truth Lagrangian constants.

Constant	Expression	Value
K_1	$2m_s ab + m_4(a + h_4)L$	0.13700 kg m ²
K_2	$2m_s b^2 + 2I_s + m_4 L^2$	0.11530 kg m ²
$M_{11,0}$	(see [1], Eq. 28)	0.48342 kg m ²
$m_{\text{eff},1}$	$(m_1 + 2m_s + m_4)a + m_4 h_4$	0.95000 kg m
$m_{\text{eff},2}$	$2m_s b + m_4 L$	0.41000 kg m

3 Methodology

3.1 Dataset Generation via CTC Simulation

Training data is produced by simulating a Computed Torque Controller (CTC) tracking sinusoidal end-effector reference trajectories of varying amplitude and frequency at a constant Cartesian speed of 5 cm/s. At each timestep the CTC uses the exact dynamics to compute the joint torques required to follow the reference, yielding physically consistent tuples $(\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}}, \mathbf{F}) \rightarrow \boldsymbol{\tau}$.

A time-varying external force $\mathbf{F}(t)$ with magnitude $|\mathbf{F}| \in [5, 15]$ N and direction $\theta \in [-90, +30]$ is applied at the end-effector. The profile is generated as a smooth random function via monotone Pchip interpolation across exponentially-spaced time nodes, guaranteeing that $|\mathbf{F}(t)|$ remains within the specified bounds without overshoot. A sensor latency of $\Delta t_F = 5$ ms is applied: the CTC receives the delayed reading $\mathbf{F}(t - \Delta t_F)$, while the same delayed value is stored as the input feature $\mathbf{F}_{\text{log}}$. This models a realistic force-sensor pipeline.

The dataset consists of:

- **Training set:** 13 792 samples from 4 trajectories (amplitude $A = 2.0$ cm, 4–7 waves).
- **OOD test set:** 13 966 samples from 4 trajectories (amplitude $A = 2.5$ cm, 4–7 waves, unseen amplitude).

- **Validation trajectory:** amplitude $A = 2.5$ cm, 9 waves — held out entirely, used only in Section 5.2.

All experiments use 10% of the training set (1 379 samples) to stress-test data efficiency. The 8-dimensional input vector is $\mathbf{x} = [q_1, q_2, \dot{q}_1, \dot{q}_2, \ddot{q}_1, \ddot{q}_2, F_x, F_y]^\top$.

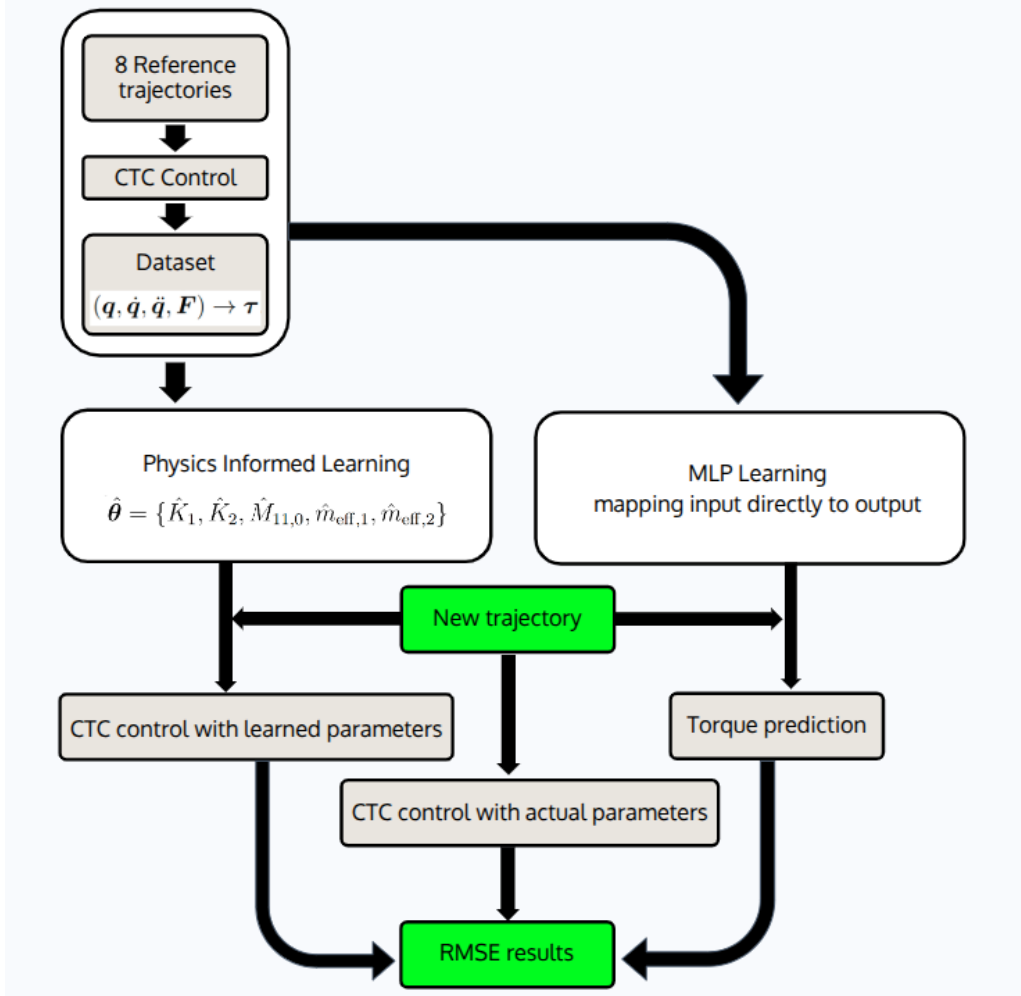


Figure 1: Overview of the methodology pipeline.

3.2 Physics-Informed Model

The physics-informed model replaces the unknown constants in (2)–(4) with five learnable scalars $\hat{\boldsymbol{\theta}} = \{\hat{K}_1, \hat{K}_2, \hat{M}_{11,0}, \hat{m}_{\text{eff},1}, \hat{m}_{\text{eff},2}\}$ and computes the predicted torque analytically:

$$\hat{\boldsymbol{\tau}} = \hat{M}(q_2) \ddot{\mathbf{q}} + \hat{C}(q_2, \dot{\mathbf{q}}) \dot{\mathbf{q}} + \hat{G}(\mathbf{q}) - Q_F(\mathbf{q}, \mathbf{F}). \quad (5)$$

The physical structure of M , C , G is hard-coded from the derivation in [1] and is never modified during training. The scalar constants are the only free parameters.

The training loss is the mean squared error on joint torques:

$$\mathcal{L}(\hat{\boldsymbol{\theta}}) = \frac{1}{N} \sum_{i=1}^N \|\hat{\boldsymbol{\tau}}(\mathbf{q}_i, \dot{\mathbf{q}}_i, \ddot{\mathbf{q}}_i, \mathbf{F}_i; \hat{\boldsymbol{\theta}}) - \boldsymbol{\tau}_i\|^2. \quad (6)$$

Optimisation uses L-BFGS (limited-memory BFGS, a quasi-Newton method) with a single full-batch closure. Because the loss landscape is smooth and well-conditioned (5 scalar parameters, analytical gradients, no activation non-linearities), convergence is achieved in fewer than 10 function evaluations.

3.3 MLP Baseline

A feedforward network with three hidden layers of 128 neurons each (35 202 parameters, ReLU activations) maps the 8-dimensional input directly to (τ_1, τ_2) . Inputs and outputs are standardised with training-set statistics. The network has no access to the mechanism’s structure.

4 Simulation Setup

4.1 Physical Parameters

The nominal physical parameters of the mechanism are listed in Tables 2 and 3. These values are used to compute the ground-truth Lagrangian constants (Table 1) and to generate the CTC training data.

Table 2: Body inertial parameters.

Symbol	Description	Value
m_1	Body 1 mass (rotating arm)	1.00 kg
I_1	Body 1 moment of inertia about G_1	0.05 kg m ²
m_s	Side-link mass (bodies 2 and 3)	0.50 kg
I_s	Side-link moment of inertia about CoM	0.01 kg m ²
m_4	Coupler mass (body 4)	1.00 kg
I_4	Coupler moment of inertia about G_4	0.01 kg m ²

Table 3: Geometric parameters.

Symbol	Description	Value
a	Distance from O to G_1 ; half-span of O_2O_3	0.30 m
b	Distance from pivot to CoM along each side link	0.13 m
c	Distance from CoM to distal end of side link	0.15 m
$L = b + c$	Total side-link length	0.28 m
e	Length of segment O_2O_3 (coupler span)	0.15 m
h_4	Perpendicular offset from midpoint of AB to G_4	0.05 m
g	Gravitational acceleration	9.81 m/s ²

4.2 Software and Integration

All simulations are implemented in Python 3.12 using PyTorch 2.x (CUDA) and SciPy. The CTC dataset is generated by integrating (1) with `scipy.integrate.solve_ivp` (RK45, $\text{rtol} = 10^{-8}$, $\text{atol} = 10^{-10}$) at a uniform timestep of $\Delta t = 10$ ms. Closed-loop validation re-integrates the same ODE with the learned parameters substituted into the CTC law.

Hardware joint limits: $q_1 \in [-90, +30]$, $q_2 \in [-50, +50]$. In the training set, q_2 has mean 0.63 rad and standard deviation 0.12 rad ($\approx 7^\circ$), reflecting small sinusoidal oscillations around the nominal configuration. The consequences of this limited excitation for parameter identifiability are analysed in Section 5.3.

5 Results

5.1 Torque Prediction Accuracy

Table 4 reports torque RMSE on the training, validation, and OOD test sets. The OOD/Train ratio measures generalisation: a value near 1.0 indicates that the model performs equally well outside the training distribution.

Table 4: Torque prediction RMSE (N·m). Training uses 1 379 samples.

Model	# Params	Train	OOD test	OOD/Train
MLP (black-box)	35 202	0.123	0.919	7.5×
Physics-informed	5	0.020	0.029	1.5×

The physics-informed model uses $7\,000\times$ fewer parameters and achieves $46\times$ lower OOD error. Its OOD performance (0.029 N·m) is better than the MLP’s within-distribution validation RMSE (0.223 N·m), demonstrating that the embedded physics structure acts as a strong inductive bias that replaces large amounts of data.

5.2 Closed-Loop Controller Validation

The quality of the learned model as a *controller* is assessed by substituting the identified parameters into a CTC and integrating the closed-loop system on the held-out 9-wave trajectory ($T = 36.9$ s). This trajectory was not used in any training or OOD dataset.

Table 5: Closed-loop tracking performance (amplitude 2.5 cm, 9 waves, $T = 36.9$ s).

Controller	RMSE	Max error	Notes
CTC (ground-truth params)	0.094 mm	0.691 mm	reference
Physics-informed CTC	0.635 mm	2.395 mm	learned params
MLP (feedforward)	613.7 mm	715.7 mm	diverges at $t = 1.3$ s

The physics-informed controller remains stable for the full 36.9 s trajectory, achieving sub-3 mm peak error. The MLP controller produces physically inconsistent torques

outside its training distribution; the closed-loop simulation diverges immediately. This confirms that a black-box model trained on inverse dynamics samples cannot be directly used as a feedforward controller without physics structure.

5.3 Physical Parameter Identification

Table 6: Identified vs. ground-truth Lagrangian constants. Init error: deviation of the initial guess from the true value.

Parameter	True	Init	Learned	Error
K_1	0.13700	0.02740	0.02740	80%
K_2	0.11530	0.02306	0.02306	80%
$M_{11,0}$	0.48342	0.14503	0.14503	70%
$m_{\text{eff},1}$	0.95000	1.18750	0.94990	0.01%
$m_{\text{eff},2}$	0.41000	0.32800	0.41000	0.00%

The effective mass parameters $m_{\text{eff},1}$ and $m_{\text{eff},2}$ are recovered with near-perfect accuracy despite starting from initial errors of 25% and 20% respectively. These constants dominate the gravity vector $G(\mathbf{q})$ and are well-excited because q_1 varies over $[-90, +30]$ throughout every trajectory, making $\cos q_1$ and $\cos(q_1 + q_2)$ strongly informative.

The constants K_1 , K_2 , and $M_{11,0}$ are not uniquely identifiable from the available data. These three constants enter $M(q_2)$ only through $\cos q_2$, which varies over a range of the workspace in the training set. In this regime $K_1 \cos q_2 \approx K_1 c_0$ for a near-constant c_0 , making K_1 and $M_{11,0}$ collinear in the regression: multiple pairs $(K_1, M_{11,0})$ produce the same torques within the operating workspace. The optimiser converges to the combination nearest to its initialisation rather than the ground truth.

Despite this partial identifiability, the learned parameter set is *functionally correct* within the operating workspace: the OOD/Train ratio of $1.5\times$ and the closed-loop tracking error of 0.635 mm confirm that the identified model generalises well and supports stable control.

6 Conclusion

This report demonstrates that embedding the Lagrangian structure of a mechanism into the learning problem yields a dramatically more data-efficient and generalisable inverse dynamics model than a black-box MLP. With only 5 learnable parameters and 1 379 training samples, the physics-informed model achieves:

- $46\times$ lower out-of-distribution torque RMSE (0.029 vs. 0.919 N·m),
- stable closed-loop control (0.635 mm tracking) on a held-out trajectory where the MLP diverges within 1.3 s,
- near-perfect recovery of the gravitational effective masses ($m_{\text{eff},1}$ to 0.01%, $m_{\text{eff},2}$ to 0.00%) from torque measurements alone.

The inertia constants K_1 , K_2 , $M_{11,0}$ remain unidentifiable from trajectories with limited q_2 excitation ($\sigma_{q_2} \approx 7$). This is a property of the experimental design: richer excitation spanning the full joint space would resolve the collinearity. The learned parameter combination is nonetheless functionally adequate for control within the operating workspace.

Future directions include residual physics-informed learning (analytical model plus neural residual correction for unmodelled effects), active trajectory design for improved identifiability, and extension to the full unknown physical parameter systems.

References

- [1] V. T. Phu, *Lagrangian Dynamic Formulation of a Planar Mechanism with a Four-Bar Linkage Mounted on a Rotating Base*, Technical Report, Paris-Saclay University MMVAI, 2025–2026.
- [2] B. Siciliano, L. Sciavicco, L. Villani, and G. Oriolo, *Robotics: Modelling, Planning and Control*. Springer, 2009.
- [3] M. Raissi, P. Perdikaris, and G. E. Karniadakis, “Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations,” *Journal of Computational Physics*, vol. 378, pp. 686–707, 2019.